

Ex. No 7: Use AFLogical OSE to Extract Data from an Android Device

Date:

Aim

To extract logical data such as contacts, SMS, and call logs from an Android device using AFLogical OSE and analyze the collected evidence for forensic purposes.

Procedure

Step 1: Extract All ZIP Files

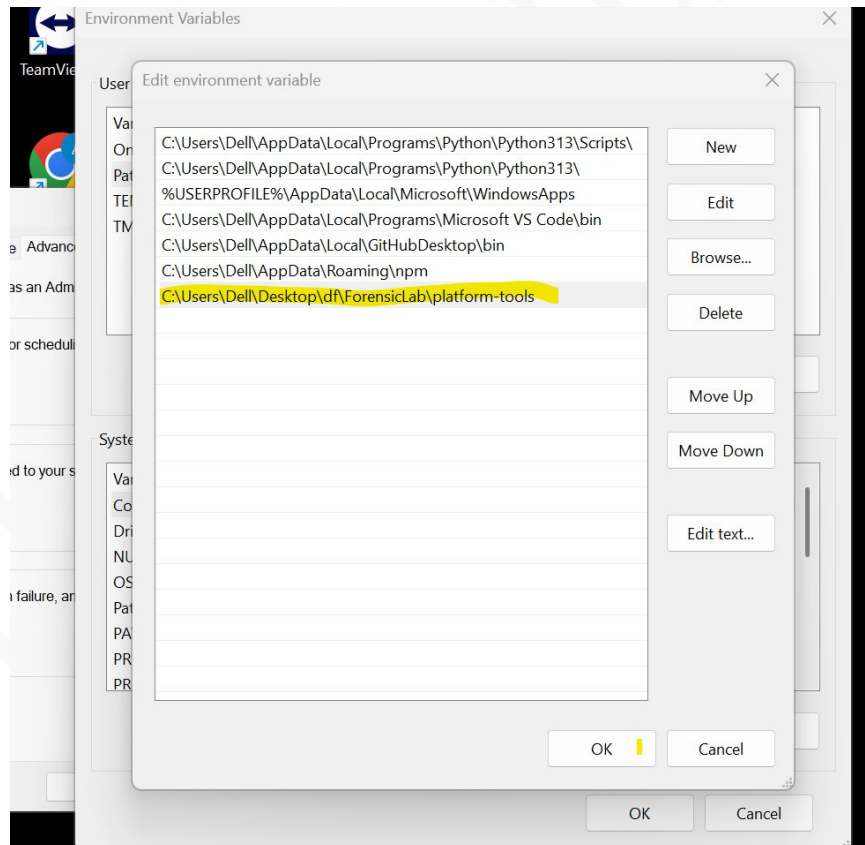
Create a main folder for your lab at C:\ForensicLab and extract the downloaded packages into it:

- **Android Platform Tools (ADB):** C:\ForensicLab\platform-tools\
- **AFLogical OSE ZIP:** C:\ForensicLab\aflogical-ose\
- **Google USB Driver (Windows):** C:\ForensicLab\usb-driver\

Note: If AFLogical OSE does not include an APK file, use Santoku Linux to extract or build it from the source ZIP.

Step 2: Add platform-tools to PATH

Open Control Panel → System → Advanced system settings → Environment Variables. Under User Variables, edit Path and append C:\ForensicLab\platform-tools so adb can be run from any directory.



Verify the installation:

```
adb version
```

```
Microsoft Windows [Version 10.0.26200.6901]
(c) Microsoft Corporation. All rights reserved.

C:\Users\Dell\Desktop\df\ForensicLab\platform-tools>adb version
Android Debug Bridge version 1.0.41
Version 36.0.0-13206524
Installed as C:\Users\Dell\Desktop\df\ForensicLab\platform-tools\adb.exe
Running on Windows 10.0.26200

C:\Users\Dell\Desktop\df\ForensicLab\platform-tools>|
```

Step 3: Install Google USB Driver (Windows)

Connect your Android phone via USB. Open Device Manager → find your phone → Update Driver → Browse my computer → Select C:\ForensicLab\usb-driver.

Verify the connection:

adb devices

```
Microsoft Windows [Version 10.0.26200.6901]
(c) Microsoft Corporation. All rights reserved.

C:\Users\Dell\Desktop\df\ForensicLab\platform-tools>adb version
Android Debug Bridge version 1.0.41
Version 36.0.0-13206524
Installed as C:\Users\Dell\Desktop\df\ForensicLab\platform-tools\adb.exe
Running on Windows 10.0.26200

C:\Users\Dell\Desktop\df\ForensicLab\platform-tools>adb devices
* daemon not running; starting now at tcp:5037
* daemon started successfully
List of devices attached
10BD541MFE001FN unauthorized

C:\Users\Dell\Desktop\df\ForensicLab\platform-tools>|
```

Step 4: Enable Developer Options on the Phone

On your phone, navigate to Settings → About phone → Tap Build number 7 times. Then go back to Settings → Developer options and enable:

- USB Debugging
- Install via USB (if available)

Step 5: Connect Phone and Check ADB Connection

Connect your phone using a high-quality data cable. In the terminal, execute 'adb devices' and accept the 'Allow USB debugging' prompt on the device screen.

Step 6: Install AFLogical on the Phone

Install the forensic extraction app (APK) onto the Android device:

```
adb install C:\ForensicLab\aflogical-ose\AFLogical-OSE.apk
```

```
C:\Users\Dell\Desktop\df\ForensicLab\platform-tools>adb install --bypass-low-target-sdk-block "C:\Users\Dell\Desktop\df\ForensicLab\AFLogical-OSE_1.5.2.apk"
Performing Streamed Install
Success
C:\Users\Dell\Desktop\df\ForensicLab\platform-tools>
```

Verification: Check your phone — the AFLogical app will appear in your application drawer.

Step 7: Run AFLogical OSE on the Phone

Open the AFLogical app on the device. Grant all requested permissions (Contacts, SMS, Storage, Call Logs). Select data to extract (Contacts, SMS, Call Logs, MMS, Calendar), and tap Start Extraction.

Default save location: /sdcard/aflogical/ or /storage/emulated/0/aflogical/.

Step 8: Copy Extracted Data to Your Computer

Pull the extracted forensic data to your analysis workstation:

```
adb pull /sdcard/aflogical C:\ForensicLab\output
```

```
C:\Users\Dell\Desktop\df\ForensicLab\platform-tools>adb install --bypass-low-target-sdk-block "C:\Users\Dell\Desktop\df\ForensicLab\AFLogical-OSE_1.5.2.apk"
Performing Streamed Install
Success

C:\Users\Dell\Desktop\df\ForensicLab\platform-tools>adb pull /sdcard/forensics/20251028.1242 "C:\Users\Dell\Desktop\df\ForensicLab\output"
/sdcard/forensics/20251028.1242/: 8 files pulled, 0 skipped. 12.1 MB/s (1744461 bytes in 0.137s)
```

Inspect the pulled CSV forensic artifacts:

```
dir C:\ForensicLab\output
```

Name	Date modified	Type	Size
 CallLog Calls	28-10-2025 12:44	Microsoft Excel Com...	703 KB
 Contacts Phones	28-10-2025 12:44	Microsoft Excel Com...	11 KB
 e024862b-b46e-4548-8b59-cb868aa86344	28-10-2025 12:44	JPG File	179 KB
 IMG_20231127_104636_01	28-10-2025 12:44	JPG File	137 KB
 info	28-10-2025 12:44	Microsoft Edge HTM...	328 KB
 MMS	28-10-2025 12:44	Microsoft Excel Com...	76 KB
 MMSParts	28-10-2025 12:44	Microsoft Excel Com...	54 KB
 SMS	28-10-2025 12:44	Microsoft Excel Com...	220 KB

Step 9: Clean Up & Safe Disconnect

Uninstall the forensic agent from the device and disconnect safely:

```
adb uninstall com.viaforensics.android.aflogical
```

Rubrics

Criteria	Mark Allotted	Mark Awarded
1. GitHub Activity & Submission Regularity	3	
2. Application of Forensic Tools & Practical Execution	3	
3. Documentation & Reporting	2	
4. Engagement, Problem-Solving & Team Collaboration	2	
Total	10	

Result

- Successfully connected the Android device to the forensic workstation using ADB.
- Installed and executed AFLogical OSE on the device for logical data extraction.
- Extracted contacts, call logs, SMS, MMS, and calendar evidence to CSV files on the workstation for forensic examination.